TSGD: The Sister of SGD

Yassine Maziane Peter Richtárik

Abstract

Stochastic gradient descent (SGD) estimates gradients. Transported SGD (TSGD) estimates their changes and uses them to update an initial gradient estimate along the optimization trajectory. Estimating changes can be much easier than estimating gradients: even when individual gradients are highly noisy, shared additive noise cancels exactly from their differences, giving exact transport. With exact initialization as well, TSGD follows gradient descent exactly, regardless of the noise magnitude. This known recursion appears in several variance-reduced methods; here we study it directly as a fundamental estimation mechanism. Its accuracy depends on two error sources: initialization error, which persists, and transport errors, which accumulate. Centered similarity controls the latter through noise variation rather than pointwise noise magnitude. We give a clean, unified convergence analysis that separates these contributions, and derive exact quadratic dynamics that quantify how initialization and transport errors interact. Simple examples and least-squares experiments identify regimes favoring TSGD or SGD. These results clarify both the value of pure transport and why modern optimizers combine it with fresh estimation.

1 Introduction

Stochastic gradient descent (SGD) estimates gradients afresh. Transported SGD (TSGD) estimates gradient changes and uses them to update an existing estimate. SGD is a cornerstone of stochastic optimization and a workhorse of machine learning: replacing the exact gradient with a stochastic estimate makes each update simple and scalable [2]. TSGD follows an equally natural principle. Starting from an initial gradient estimate, it estimates the change between consecutive iterates and adds that change to its current estimate, transporting gradient information along the optimization trajectory.

Estimating changes can be much easier than estimating absolute quantities. Two stochastic gradients evaluated with the same sample can share a noise component; this shared component cancels when they are subtracted, while noise that differs between the two evaluations may remain. In particular, additive noise that is shared across evaluations cancels exactly from their gradient difference, even when individual gradients are highly noisy. The difference then gives the exact gradient change. With exact initialization as well, TSGD follows gradient descent exactly, regardless of the noise magnitude. This is a pathwise identity, not merely a statement in expectation.

The two methods therefore depend on different aspects of the noise. Classical SGD guarantees depend on its pointwise magnitude, commonly bounded by a variance parameter σ^2 [7]. TSGD guarantees, on the other hand, depend on how the noise varies between the coupled evaluations along the trajectory. Centered δ-similarity quantifies this variation: small δ means that stochastic gradient changes closely match deterministic gradient changes. Thus, transport can remain accurate even when pointwise noise is large. We stress that centered similarity applies to population objectives as well as finite sums.

Two sources of error determine TSGD's accuracy: error in the initial gradient estimate and error in the estimated gradient changes. The initial error persists; new transport errors accumulate

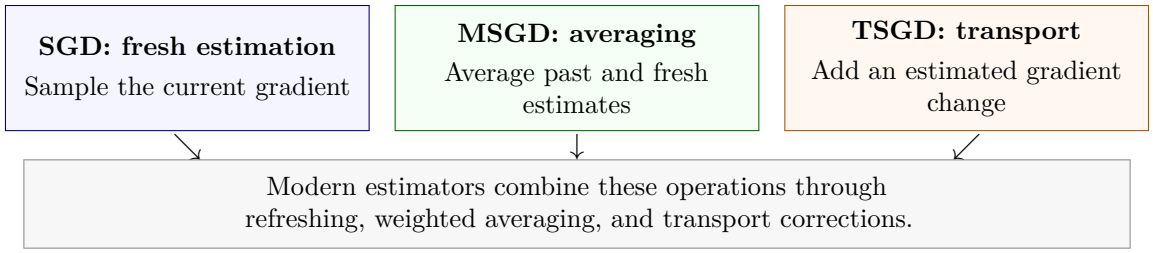


Figure 1: Three fundamental estimation mechanisms. The arrows indicate complementary operations available to modern optimizers, not a requirement that every method use all three.

along the trajectory. Accurate differences preserve a good initial estimate, but cannot correct an inaccurate one. The TSGD guarantee therefore depends on both δ and the accuracy of the initial estimate. At fixed values of these quantities, it is independent of σ, although the noise magnitude can affect the cost of obtaining an accurate initial estimate.

Least-squares regression provides a concrete example. Label noise enters the stochastic gradient $u(u^\top x - b)$, but cancels from the shared-sample difference $uu^\top(x-y)$. It therefore affects fresh gradient estimates and TSGD's initialization, but not its subsequent transport increments (Section 2.2).

The transport recursion is not new. It appears in established variance-reduced algorithms such as SARAH, SPIDER, and PAGE [5, 13, 14], and independent single-run analyses already exist [15, 16, 19]. It is commonly encountered as part of methods that combine transport with refreshing or mixing. Our purpose is to study the mechanism directly, making its strengths and limitations as explicit as those of SGD.

Momentum SGD (MSGD) provides a third fundamental mechanism: averaging fresh gradients with past estimates. Section 2 introduces all three estimators, and Section 5 describes how modern optimizers use TSGD and combine transport with fresh gradients and momentum averaging.

The paper develops this viewpoint through three contributions:

- **Fundamental viewpoint.** We present TSGD alongside SGD as a basic way of using stochastic-gradient information: estimating changes rather than absolute gradients. Similarity explains when these changes can remain accurate despite noisy individual gradients.
- **Theory and understanding.** We explain similarity, separate initialization and transport errors, and give a finite-horizon guarantee under conditional assumptions on the gradient change increments. Exact quadratic dynamics quantify their interaction and show how the estimator can vanish while the true gradient does not.
- **Comparison and connections.** We construct oracles with identical pointwise gradient distributions but different transport accuracy, and give exact examples showing when SGD or TSGD has the smaller stationarity error. Least-squares experiments illustrate these regimes and the effect of initialization. We compare convergence guarantees and oracle costs, and identify how established optimizers combine transport with fresh estimation and momentum averaging.

The examples precede the convergence analysis so that the role of each assumption is visible before the theorem. Detailed proofs and additional calculations appear in the appendices.

2 Three Fundamental Estimators and Similarity

2.1 SGD, MSGD, and TSGD

After taking a step to x_{k+1}, we need a gradient estimate to determine the next update. The identity

$$\nabla f(x_{k+1}) = \nabla f(x_k) + [\nabla f(x_{k+1}) - \nabla f(x_k)]$$

suggests two approaches. SGD estimates the left-hand side directly by drawing a fresh stochastic gradient at the new point. TSGD estimates the right-hand side: its current estimate g_k already approximates the first term, so it needs new information only for the change. Using a stochastic gradient oracle h with the same fresh sample at both endpoints gives

$$g_{k+1} = \underbrace{g_k}_{\approx \nabla f(x_k)} + \underbrace{h(x_{k+1}, \xi_{k+1}) - h(x_k, \xi_{k+1})}_{\approx \nabla f(x_{k+1}) - \nabla f(x_k)}.$$

Thus TSGD updates the gradient information it already has. The sample ξ_{k+1} is shared within the difference and drawn afresh at each iteration. For a finite sum, this means evaluating the same randomly selected component at both points. TSGD retains the information in g_k, including its estimation error; the effect of these inherited errors is examined in Section 2.3.

We formalize the setup as follows.

Assumption A1 (Smooth, lower-bounded objective)**.** The function $f : \mathbb{R}^d \to \mathbb{R}$ is lower bounded, $f_{\inf} := \inf_x f(x) > -\infty$, and has an L-Lipschitz gradient with $L > 0$:

$$\|\nabla f(x) - \nabla f(y)\| \le L\|x - y\| \qquad \text{for all } x, y \in \mathbb{R}^d.$$

Assumption A2 (Unbiased stochastic gradients)**.** The oracle satisfies $\mathbb{E}_\xi[h(x, \xi)] = \nabla f(x)$ for every x. Each new sample is independent of the preceding history.

All three methods use positive deterministic stepsizes η_k and the same iterate update, starting from a deterministic x_0:

$$x_{k+1} = x_k - \eta_k g_k. \tag{2.1}$$

Their gradient estimates differ:

$$\text{SGD}: \quad g_{k+1} = h(x_{k+1}, \xi_{k+1}), \tag{2.2}$$

$$\text{MSGD}: \quad g_{k+1} = (1 - \beta_k)g_k + \beta_k h(x_{k+1}, \xi_{k+1}), \quad 0 < \beta_k \le 1, \tag{2.3}$$

$$\text{TSGD}: \quad g_{k+1} = g_k + D_{k+1}. \tag{2.4}$$

In the standard pointwise-oracle construction,

$$D_{k+1} = h(x_{k+1}, \xi_{k+1}) - h(x_k, \xi_{k+1}). \tag{2.5}$$

SGD replaces its estimate with a fresh stochastic gradient at every iteration. MSGD averages fresh and past estimates to reduce noise, but can lag behind a changing gradient. TSGD accounts for the gradient change through transport. Unless specified otherwise, SGD and MSGD initialize with $g_0 = h(x_0, \xi_0)$. TSGD permits any g_0 with finite mean-square error, including a biased estimate.

2.2 Understanding Similarity

SGD is typically analyzed under a bounded-variance assumption. Write $r(x,\xi) = h(x,\xi) - \nabla f(x)$ for the gradient noise.

Assumption A3 (Bounded variance (BV))**.** For some $\sigma \geq 0$,

$$\mathbb{E}_\xi[\|r(x,\xi)\|^2] \leq \sigma^2 \qquad \text{for all } x. \tag{2.6}$$

Transport instead uses a condition on gradient differences.

Assumption A4 (Centered δ-similarity)**.** For some $\delta \geq 0$,

$$\mathbb{E}_\xi[\|r(x,\xi) - r(y,\xi)\|^2] \leq \delta^2\|x-y\|^2 \qquad \text{for all } x, y. \tag{2.7}$$

Under uniform finite-sum sampling, this condition coincides with the Hessian-variance condition of Szlendak et al. [20] and the second-order similarity assumption of Khaled and Jin [11]. More recently, Shulgin et al. [18] generalized it through γ-similarity, which recovers this condition at $\gamma = 1$. We use the name *centered δ-similarity* to emphasize that it measures the variation of gradient noise rather than its magnitude.

Here σ bounds the magnitude of individual gradient errors, whereas δ measures their variation under shared sampling. A small δ means that stochastic gradient differences closely match deterministic gradient differences; it does not require the gradients themselves, or their changes, to be small. Similarity is a condition on the stochastic oracle and applies to both finite-sum and population objectives.

For a finite sum $f = \frac{1}{n}\sum_{i=1}^n f_i$ with quadratic components $f_i(x) = \frac{1}{2}x^\top A_i x + b_i^\top x + c_i$ and symmetric A_i, the linear terms $b_i^\top x$ cancel from gradient differences, so identical Hessians give exact transport even when individual gradients differ greatly. Hessian variation controls δ, whereas L concerns the average Hessian; neither must be smaller than the other. Appendix C.2 gives the calculation.

Scalar least squares separates the two quantities. Let $f(x) = \frac{1}{2}\mathbb{E}[(ux-b)^2]$, with feature u satisfying $0 < \mathbb{E}[u^2]$ and $\mathbb{E}[u^4] < \infty$, and label $b = ux^\star + \varepsilon$. The label noise ε is independent of u, with $\mathbb{E}[\varepsilon] = 0$ and $\mathbb{E}[\varepsilon^2] = s^2 < \infty$. The stochastic gradient is $h(x;u,\varepsilon) = u^2(x-x^\star) - u\varepsilon$, so

$$\begin{aligned} h(x;u,\varepsilon) - h(y;u,\varepsilon) &= u^2(x-y), \\ \mathbb{E}[|h(x;u,\varepsilon) - \nabla f(x)|^2] &= \delta^2(x-x^\star)^2 + Ls^2, \end{aligned} \tag{2.8}$$

where $\delta^2 = \mathrm{Var}(u^2)$ and $L = \mathbb{E}[u^2]$. Label noise increases the variance of fresh gradients but cancels from differences; feature variability sets δ. Increasing s therefore affects TSGD's sampled initialization but not its later increments. When $\delta > 0$, the pointwise noise variance is unbounded in x, so similarity holds but global BV fails. Sections 3.1 and 3.3 illustrate the regimes of exact transport ($\delta = 0$) and noiseless labels ($s = 0$), respectively.

SGD can also be analyzed under the ABC condition, which generalizes BV [12]. Both constrain the stochastic gradient at each point separately; centered similarity instead constrains noise differences at two points under shared sampling. Appendix C.1 compares these conditions.

2.3 Two Fundamental Sources of TSGD Error

The introduction identified two sources of TSGD error: initialization and transport. We now make this decomposition precise. Define $e_k = g_k - \nabla f(x_k)$, $E_k = \mathbb{E}[\|e_k\|^2]$, and $d_{k+1} = D_{k+1} - [\nabla f(x_{k+1}) - \nabla f(x_k)]$. The transport update gives

$$\boxed{e_{k+1} = e_k + d_{k+1}.} \tag{2.9}$$

Let $\mathcal{F}_k$ denote the history before drawing D_{k+1}, including x_{k+1} and g_k. Conditionally unbiased increments give

$$\mathbb{E}[g_{k+1} \mid \mathcal{F}_k] = \nabla f(x_{k+1}) + e_k.$$

Unlike SGD, TSGD is therefore generally conditionally biased: the new increment is unbiased, but the estimator inherits its previous error. The increment error d_{k+1} has zero conditional mean and is orthogonal in expectation to previous errors. Consequently,

$$E_k = E_0 + \sum_{t=0}^{k-1} \mathbb{E}[\|d_{t+1}\|^2]. \tag{2.10}$$

Even if $\mathbb{E}[e_0] = 0$, so that $\mathbb{E}[e_k] = 0$, the mean-square error E_k cannot decrease. Initialization error persists, while transport errors accumulate. Similarity bounds each contribution by the squared displacement, yielding

$$E_k \le E_0 + \delta^2 \sum_{t=0}^{k-1} \eta_t^2 \mathbb{E}[\|g_t\|^2]. \tag{2.11}$$

This explains both the opportunity and the difficulty: accurate changes preserve useful information, but even small errors can accumulate. Section 4 states the conditional assumptions directly on D_{k+1}, allowing transport beyond pointwise gradient oracles.

3 When Does TSGD Shine?

Through explicit examples, we now examine when TSGD benefits from accurate gradient changes and when SGD benefits from fresh estimates. We measure stationarity by the expected squared true-gradient norm, $G_k := \mathbb{E}[\|\nabla f(x_k)\|^2]$, and show how initialization quality and noise variation affect the comparison.

Alongside exact calculations, we use a vector least-squares model $f(x) = \frac{1}{2}\mathbb{E}[\|Ax - b\|^2]$ as a running experiment. Its population Hessian is held fixed while feature variability sets δ and label noise varies independently (Section 2.2).

3.1 Accurate Differences Despite Noisy Gradients

Consider $f(x) = x^2/2$ and an unbiased oracle $h(x, \xi) = x + \xi$, where $\mathbb{E}[\xi] = 0$ and $\mathbb{E}[\xi^2] = \sigma^2$. The individual gradients can be arbitrarily noisy, yet

$$h(x, \xi) - h(y, \xi) = x - y = \nabla f(x) - \nabla f(y).$$

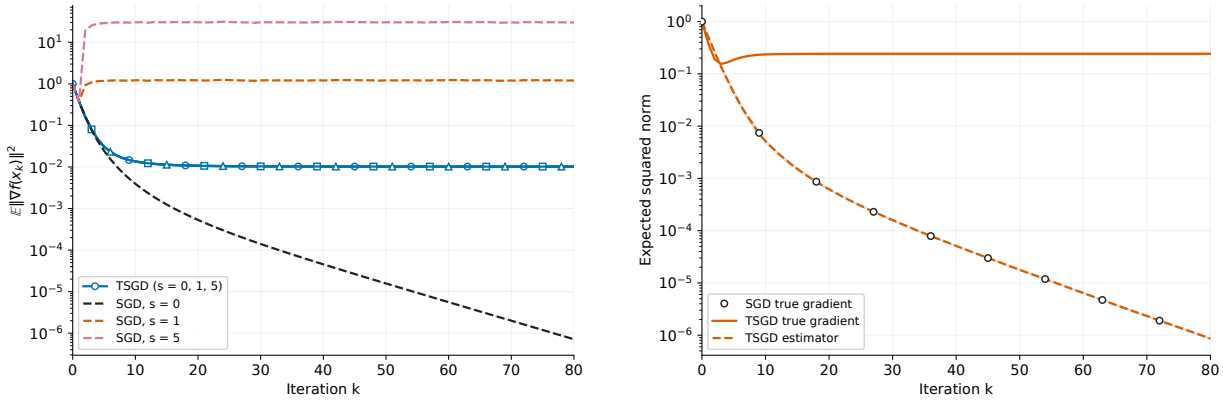
Thus $\delta = 0$, independently of σ. With exact initialization, TSGD follows gradient descent exactly. Indeed, $g_k = \nabla f(x_k)$ and an exact difference give

$$g_{k+1} = g_k + \nabla f(x_{k+1}) - \nabla f(x_k) = \nabla f(x_{k+1}).$$

Induction from $g_0 = \nabla f(x_0)$ proves this for any objective. Appendix B.2 gives the resulting GD convergence guarantee and proves that the gradients tend to zero.

For the present quadratic, take $x_0 = 1$, $\eta = 1$, and any initial estimate $g_0 = 1 + e_0$ with $E_0 = \mathbb{E}[e_0^2] < \infty$, biased or not. Then $x_1 = -e_0$ and $g_1 = g_0 + (x_1 - x_0) = 0$, so TSGD stays at $x_k = -e_0$ for all $k \ge 1$. SGD instead uses a fresh gradient at every update, giving $x_{k+1} = -\xi_k$. Consequently,

$$\boxed{G_k^{\mathrm{TSGD}} = E_0, \qquad G_k^{\mathrm{SGD}} = \sigma^2, \qquad k \ge 1.}$$



(a) Label noise cancels from transport.

(b) False convergence of the transported estimate.

Figure 2: Least-squares experiments with a common stepsize. (a) With exact initialization and matched feature draws, TSGD's trajectories coincide across label-noise levels, while SGD's error floor grows with the noise. (b) With noiseless labels and variable features, SGD's true gradient decays to zero, while TSGD's estimator vanishes but its true gradient does not. Curves show means over 4,000 trajectories with pointwise bands of ± 2 standard errors; the appendix gives the full setup.

Transport preserves the quality of its initial information, good or bad: TSGD has the smaller error when $E_0 < \sigma^2$ and the larger error when $E_0 > \sigma^2$. A single exact gradient evaluation at initialization, $g_0 = \nabla f(x_0)$, gives $E_0 = 0$: TSGD reaches the minimizer in one step and $G_k^{\mathrm{TSGD}} = 0$ for every $k \geq 1$, without any further exact gradient evaluations. In contrast, single-sample initialization $g_0 = h(x_0, \xi_0)$ gives the same error as SGD.

Figure 2(a) illustrates the same separation with varying features. Holding the initial estimate and feature draws fixed, TSGD's trajectories coincide across label-noise levels, while SGD's error floor increases with the label noise. TSGD nevertheless plateaus because $\delta > 0$ and transport errors accumulate at a constant stepsize; without label noise, SGD eventually overtakes it, as the fresh-gradient example in Section 3.3 also illustrates.

3.2 Fixed Noise Magnitude, Variable Similarity

To determine whether noise magnitude alone explains the relative behavior of SGD and TSGD, we compare oracles with the same pointwise noise distribution but different noise variation. For $\omega \geq 0$, consider

$$f(x) = \frac{L}{2}x^2, \qquad h(x, \varphi) = Lx + \sqrt{2}\sigma \cos(\omega x + \varphi), \qquad \varphi \sim \mathrm{Unif}[0, 2\pi). \tag{3.1}$$

Each new sample is an independent phase; a transport difference uses the same phase at both endpoints. For the noise $r(x, \varphi) = h(x, \varphi) - Lx$,

$$\mathbb{E}[r(x, \varphi)^2] = \sigma^2, \qquad \mathbb{E}[|r(x, \varphi) - r(y, \varphi)|^2] = 4\sigma^2 \sin^2\left(\frac{\omega(x - y)}{2}\right). \tag{3.2}$$

The smallest global similarity constant is $\delta = \sigma\omega$ (Appendix D.4). Pointwise noise has the same distribution for every x and ω, so SGD has identical expected squared-gradient dynamics at every frequency. TSGD, however, has exact transport at $\omega = 0$, while its similarity constant grows as $\delta = \sigma\omega$. Its actual transport error depends on the displacement through the sine term.

Simulations in Appendix D.5 (Figure 4) confirm this distinction. At small frequencies, TSGD retains accurate information despite persistent pointwise noise; at $\omega = 2$, its error exceeds SGD's.

Initialization	Exact transport	Stochastic transport
Exact	Gradient descent	Accumulated transport error
Inexact	Persistent initial error	Both errors and their interaction

Table 1: Four regimes in the quadratic model, with a stable constant stepsize. The exact-transport identities also hold for general objectives; the explicit limiting errors and stability condition are established in Appendix B.

Thus, at the same pointwise noise level and stepsize, changing the noise variation changes which method performs better.

3.3 When Fresh Gradients Are Preferable

Now let

$$f(x) = \frac{x^2}{2}, \qquad h(x,a) = ax, \qquad \mathbb{P}(a=0) = \mathbb{P}(a=2) = \frac{1}{2}, \qquad x_0 = 1. \tag{3.3}$$

Here the gradient-noise variance is x^2: fresh gradients become exact at the minimizer, although global BV fails. Similarity holds with $\delta = 1$. At the common stepsize $\eta = 1/2$, each SGD sample either leaves the iterate unchanged or sends it to zero, so $G_k^{\mathrm{SGD}} = 2^{-k}$. In contrast, exactly initialized TSGD satisfies $g_{k+1} = (1 - a_{k+1}/2)g_k$. Its estimator becomes zero at the first sample $a_{k+1} = 2$, wherever the iterate happens to be. The exact calculation in Appendix D.2 gives

$$g_k \longrightarrow 0 \quad \text{almost surely}, \qquad G_k^{\mathrm{TSGD}} \longrightarrow \frac{1}{2}.$$

This is false convergence: the estimator tends to zero while the expected squared norm of the true gradient stays positive. Fresh information can therefore be preferable even when initialization is exact.

Figure 2(b) shows the same phenomenon on least squares with noiseless labels and highly variable features. SGD's true gradient decays in mean square, as does TSGD's estimator, but TSGD's expected squared true-gradient norm remains positive. A small transported estimate therefore does not certify a small true gradient.

3.4 What Transport Cannot Do

To separate the effects of initialization and transport, consider $f(x) = \lambda x^2/2$ and $h(x;a,b) = (\lambda + a)x + b$, with $\lambda > 0$, independent noises satisfying $\mathbb{E}[a] = \mathbb{E}[b] = 0$, $\mathbb{E}[a^2] = \delta^2$, and $\mathbb{E}[b^2] < \infty$. Shared sampling cancels b. Table 1 summarizes the four cases, whose exact dynamics are developed in Appendix B.

Exact transport preserves initialization error. The behavior in Section 3.1 extends beyond quadratics. When $\delta = 0$, the error identity gives $e_k = e_0$, hence $g_k = \nabla f(x_k) + e_0$ for every k. With $e_0 = 0$ this is GD; otherwise it is GD on $f(x) + \langle e_0, x\rangle$. The reason is that adding a linear term leaves all gradient differences unchanged, so differences alone cannot recover a missing constant gradient offset (Appendix B.3).

Stochastic transport can produce a nonzero error floor. For $0 < \eta < 2\lambda/(\lambda^2 + \delta^2)$ and $M_0 := \mathbb{E}[|g_0|^2]$, the exact quadratic dynamics give

$$\lim_{k\to\infty} \mathbb{E}[|g_k|^2] = 0, \qquad \lim_{k\to\infty} G_k = E_0 + \frac{\eta\delta^2 M_0}{2\lambda - \eta(\lambda^2 + \delta^2)}. \tag{3.4}$$

With exact initialization, $M_0 = \lambda^2 x_0^2$: if $\delta > 0$ and $x_0 \neq 0$, the estimate vanishes in mean square while the true gradient has a positive limiting mean-square norm. Appendix B.4 proves both limits.

The two errors interact. Initialization affects the residual directly through E_0 and indirectly through M_0, which controls the steps and the transport errors they generate. For centered initialization error, $M_0 = \lambda^2 x_0^2 + E_0$, so improving the initial estimate reduces both contributions. Appendix B.5 derives this interaction and the associated conservation law. Appendix B.6 varies initialization accuracy and similarity on least squares and compares the observed residuals with their exact limits.

4 Convergence Theory and Comparison with SGD

The examples distinguish noise magnitude from noise variation. We now make this distinction explicit in finite-horizon stationarity guarantees. Using G_k from the preceding section, write

$$\Delta_0 := f(x_0) - f_{\inf}, \qquad \overline{G}_K := \frac{1}{K} \sum_{k=0}^{K-1} G_k,$$

with a superscript identifying the algorithm when needed.

4.1 Main Convergence Guarantee for TSGD

Using the history $\mathcal{F}_k$ defined in Section 2.3, write $m_k := \nabla f(x_{k+1}) - \nabla f(x_k)$. We impose two conditions on the increments actually used along the trajectory.

Assumption A5 (Unbiased transport)**.** The increments satisfy

$$\mathbb{E}[D_{k+1} \mid \mathcal{F}_k] = m_k. \tag{4.1}$$

Assumption A6 (Similarity along the trajectory)**.** For some $\delta \geq 0$,

$$\mathbb{E}[\|D_{k+1} - m_k\|^2 \mid \mathcal{F}_k] \leq \delta^2 \|x_{k+1} - x_k\|^2. \tag{4.2}$$

Assumptions A2 and A4 imply these two conditions for shared-sample gradient differences. The conditional similarity requirement may be weaker than global similarity because it concerns only the generated increments. Stating them directly on the increments also allows transport without a pointwise gradient oracle.

Theorem 4.1 (Finite-horizon stationarity of TSGD)**.** *Suppose the objective is smooth and lower bounded (Assumption A1), and the transport increments are conditionally unbiased and satisfy centered δ-similarity (Assumptions A5–A6). Allow any initial estimate with $E_0 < \infty$, including biased ones. For an integer $K \geq 1$, choose a constant stepsize $\eta > 0$ satisfying*

$$L\eta + (K-1)\delta^2\eta^2 \leq 1. \tag{4.3}$$

Then

$$\overline{G}_K^{\mathrm{TSGD}} \leq \frac{2\Delta_0}{\eta K} + E_0. \tag{4.4}$$

Why the guarantee holds. Let $F_k := \mathbb{E}[f(x_k) - f_{\inf}]$ and $M_k := \mathbb{E}[\|g_k\|^2]$. Smoothness gives

$$F_{k+1} \le F_k - \frac{\eta}{2} G_k - \frac{\eta}{2}(1 - L\eta) M_k + \frac{\eta}{2} E_k. \tag{4.5}$$

The error identity (2.10) and similarity imply $E_k \le E_0 + \delta^2 \eta^2 \sum_{t<k} M_t$. When the descent inequalities are summed, an error generated at step t enters at most $K - 1 - t$ later estimates. Condition (4.3) ensures that its total contribution is no larger than the corresponding negative M_t term. This leaves the initialization error and the initial objective gap in (4.4). Appendix A gives the full proof, retaining the terminal objective gap and all estimator-norm terms. Neither bounded variance nor a pointwise gradient oracle is required.

Choosing the largest admissible stepsize gives

$$\boxed{\overline{G}_K^{\mathrm{TSGD}} \le \frac{2L\Delta_0}{K} + \frac{2\delta\Delta_0}{\sqrt{K}} + E_0.} \tag{4.6}$$

The TSGD bound does not explicitly depend on σ, but this dependence may enter through E_0. For example, initializing g_0 by averaging B_0 independent unbiased stochastic gradients at x_0 gives $E_0 \le \sigma^2/B_0$ under bounded variance. Figure 5 illustrates how a more accurate sampled initial estimate lowers TSGD's residual error. Thus, TSGD's sensitivity to gradient-noise magnitude depends on how its initial estimate is constructed.

The terms correspond to deterministic optimization, accumulated transport noise, and initialization error. Small δ permits larger steps and a smaller transport contribution. With $\delta = 0$, the admissible step is $1/L$; if also $E_0 = 0$, we recover gradient descent. The additive E_0 term reflects the error that pure transport preserves, and its coefficient is sharp (Appendix B). The guarantee approaches E_0 as the horizon grows. This floor can be attained even with exact transport, as Section 3.4 shows. Stochastic transport can create an additional floor when the stepsize is held fixed. For $\delta > 0$, the theorem controls this accumulation by choosing a smaller stepsize for a longer run.

Related single-run guarantees appear in SARAH-IN, iSARAH-IN, and PAGE-NS [15, 16, 19]. The formulation here expresses the argument directly through unbiased increments and their centered similarity.

4.2 Comparison with SGD

For SGD, unbiasedness and BV give the standard smooth nonconvex guarantee [7]. The fixed step $\eta = 1/L$ yields $\overline{G}_K^{\mathrm{SGD}} \le 2L\Delta_0/K + \sigma^2$. For $\Delta_0 > 0$, optimizing the constant step for the prescribed horizon replaces this persistent variance term by a decaying one, as shown in Table 2. The exact stepsizes and derivations are in Appendix A.

The comparison reflects the different information the methods use: SGD's bound depends on noise magnitude, while TSGD's depends on noise variation and initial accuracy. Small δ can favor transport even when σ is large. Neither BV nor the more general ABC condition is comparable with global similarity (Appendix C.1), so these bounds apply under different assumptions and do not establish universal superiority.

4.3 Computational Considerations

TSGD requires an initial estimate and, in the shared pointwise-oracle implementation, two gradient evaluations per transport increment. Accurate differences may nevertheless need far fewer samples than accurate absolute gradients: small-batch transport can preserve information obtained from a more expensive initial estimate. The benefit depends on both its initial cost and the subsequent

	SGD	TSGD
Noise assumption	BV: pointwise magnitude σ	Centered similarity: variation δ
Tuned bound on $\overline{G}_K$	$\frac{2L\Delta_0}{K} + \sqrt{\frac{2L\Delta_0\sigma^2}{K}}$	$\frac{2L\Delta_0}{K} + \frac{2\delta\Delta_0}{\sqrt{K}} + E_0$
Pointwise evaluations	K	$B_0 + 2(K-1)$

Table 2: Finite-horizon guarantees under each method's noise assumptions, with constant stepsizes optimized separately. Both require a smooth, lower-bounded objective. Costs count K iterate updates and an initialization using B_0 evaluations for TSGD.

accuracy of the differences. Iteration counts alone therefore do not compare computational work. Appendix E gives batch initialization costs and comparisons at equal oracle budgets.

5 TSGD as a Building Block of Modern Optimizers

5.1 Transport Is Already Everywhere

Many modern stochastic optimizers combine fresh gradients with transported estimates. Their common building blocks are the three operations in Figure 1: fresh estimation, averaging, and transport. Write H_{k+1} for a fresh unbiased estimate of $\nabla f(x_{k+1})$, D_{k+1} for an estimate of its change from x_k, and $g_k + D_{k+1}$ for the transported estimate. A stochastic gradient difference uses the same fresh sample or minibatch at both endpoints. Table 3 summarizes the resulting estimator relationships. Here $\alpha, \beta \in [0,1]$ weight fresh estimates, and γ scales the MARS transport correction.

Method	Estimator at iteration $k+1$	Role of transport
SARAH / SPIDER	H_{k+1} or $g_k + D_{k+1}$	Transport between periodic refreshes.
Geom-SARAH	H_{k+1} or $g_k + D_{k+1}$	Transport for a geometric number of steps.
PAGE	H_{k+1} or $g_k + D_{k+1}$	Refresh with probability p.
MARINA	$\nabla f(x_{k+1})$ or $g_k + D^{\mathcal{Q}}_{k+1}$	Communicate compressed gradient changes.
STORM / MVR	$(1-\alpha)(g_k + D_{k+1}) + \alpha H_{k+1}$	Average fresh and transported estimates.
MARS (unclipped)	$(1-\beta)(g_k + \gamma D_{k+1}) + \beta H_{k+1}$	Scale the transport correction within momentum.

Table 3: Transport in modern optimizers. Rows with two alternatives select one branch according to a restart or refresh rule; the last two rows form weighted combinations. $D^{\mathcal{Q}}_{k+1}$ denotes an unbiased compressed increment. The identities describe estimators; batch sizes, stepsizes, and iterate updates belong to each algorithm.

SVRG [10] and SAGA [4] also use stochastic gradient differences to reduce variance. SVRG compares component gradients at the current iterate and an anchor point held fixed within each

epoch, adding the full gradient at that anchor point. SAGA instead stores past gradients for individual components and combines the sampled component's gradient difference with the average of the stored gradients, updating one entry at each iteration. Both exploit noise cancellation in gradient differences. Unlike TSGD, they do not transport the previous gradient estimate along successive iterates: each estimate combines a current component gradient with stored reference information. In their standard forms, these estimates are conditionally unbiased, whereas pure TSGD inherits and accumulates tracking errors.

SARAH and SPIDER transport an estimate between full-gradient or large-batch recomputations [5, 14, 15]. Inexact SARAH allows batch initialization [16], while Geom-SARAH randomizes the phase length [9]. PAGE refreshes probabilistically [13]; MARINA combines exact refreshes with compressed gradient differences [8].

MVR/STORM instead averages fresh and transported estimates at each iteration [3]. The unclipped MARS estimator [21] includes MSGD at $\gamma = 0$ and MVR at $\gamma = 1$. These identities concern the stated sampling: cached gradients, clipping, and preconditioning require separate treatment. Sampling also affects the noise covariance, as emphasized by the VENUS construction [1]. Appendix E.4 gives the precise identities, error recursions, and special cases.

Gradient differences also appear in decentralized gradient tracking [17], where agents combine changes in their local gradients with information from neighboring agents to track the average gradient across the network.

Neural-network applications. Transport-based methods have been used to train both image classifiers and language models. SARAH and SARAH+ were evaluated on networks with one fully connected hidden layer on MNIST and CIFAR-10 [15]; PAGE was evaluated on LeNet, VGG, and ResNet architectures [13]; and MARS was evaluated on GPT-2 language models [21]. These experiments concern composite optimizers, not uninterrupted pure TSGD: SARAH variants restart their transport phases, PAGE refreshes probabilistically, and MARS combines gradient-difference corrections with momentum and preconditioning. In particular, although the nonconvex SARAH paper analyzes the pure-transport inner loop SARAH-IN, its neural-network experiments use restarted SARAH/SARAH+. Some practical MARS variants also approximate the correction using gradients computed with different samples, rather than the shared-sample increment studied here [21]. These applications motivate studying transport as a building block; they do not establish the performance of pure TSGD in isolation.

5.2 Why Combine Fundamental Estimators?

SGD, MSGD, and TSGD use fresh information, averaging, and transport, respectively. Modern optimizers combine them through weighted averages or alternate between fresh estimation and transport. Three motivations explain why these combinations can be useful.

Computational efficiency. An accurate absolute gradient may require a large batch, whereas a small batch can suffice for an accurate gradient change. Even with two evaluations per sample, transport can then cost much less. SARAH, SPIDER, and PAGE preserve information from an expensive gradient estimate through many cheaper transport steps. Refreshing more often limits accumulated error but reduces this saving.

Complementary strengths. Momentum averages past gradient estimates to reduce noise, but its estimate can lag behind the current gradient, especially when the gradient changes rapidly. Transport accounts for the change in the true gradient, reducing tracking lag while retaining the benefit of averaging. Fresh gradients supply absolute information that pure transport cannot recover: mixing them in damps initialization and accumulated transport errors, while introducing new

pointwise noise. MVR, STORM, and MARS balance these roles through their mixing weights and transport corrections. This complementarity matters even when the noise regime remains fixed.

Changing regimes. The relative accuracy of fresh and transported estimates can change during a single run. Similarity bounds each new transport error in proportion to the squared displacement, so shorter steps tighten this bound without removing previously accumulated error. Section 3.2 illustrates this displacement dependence even when pointwise noise variance stays constant. Fresh gradients may also become more reliable: in Section 3.3, their noise vanishes near the minimizer, while pure transport can retain a nonzero tracking error. Thus the balance can change as iterates move, steps shrink, and errors accumulate.

Accurate gradient differences can preserve a reliable estimate for many iterations. As tracking error accumulates, fresh gradients can help correct it. Choosing the mixing weights or refresh frequency means balancing this benefit against the cost and noise of fresh estimation. The difficulty is that these estimation errors are generally unknown during a run.

Improved complexity through refreshing. When bounded variance and centered similarity hold together, combining estimators can improve the available convergence guarantees on the same problem class. SGD repeatedly obtains fresh gradient information but does not explicitly exploit accurate shared-sample gradient differences. Pure TSGD exploits these differences, but obtains absolute-gradient information only through initialization. It retains initialization error and accumulates transport noise.

SPIDER [5] and PAGE [13] combine transport with periodic or probabilistic refreshing. An accurate gradient estimate can then support many inexpensive transport steps, while each refresh replaces the inherited error with the error of a fresh estimate. This limits how long transport errors accumulate and allows larger steps without requiring an expensive gradient estimate at every iteration. With suitable batch sizes and refresh schedules, these methods attain $O(\varepsilon^{-3})$ oracle complexity, compared with the $O(\varepsilon^{-4})$ guarantees for SGD and pure TSGD in Section 4. This ε^{-3} dependence matches the lower bound of Fradin et al. [6] for zero-respecting algorithms on this same problem class; Appendix E.3 gives the precise statement. The comparison shows how combining the mechanisms can improve a guarantee; it does not establish an $\Omega(\varepsilon^{-4})$ lower bound for pure TSGD.

6 Conclusion

Transported SGD is a fundamental way to use stochastic-gradient information: it estimates changes and carries an existing estimate along the trajectory. With a sufficiently accurate initial estimate, small variation in gradient noise can make transport effective even when individual gradients are very noisy. Its performance depends on the initialization error it retains and the transport errors it accumulates. The recursion is already studied on its own in methods such as SARAH-IN, and also appears within more elaborate optimizers. Our presentation emphasizes its role as a fundamental estimator and makes its two error sources explicit. We also briefly discuss reasons for combining transport with fresh estimation and momentum averaging.

References

[1] Anonymous authors. VENUS: Decoupling the samples in MARS benefits small-batch training. Manuscript under review at ICLR 2027; version supplied by the authors, 2026.

[2] Léon Bottou, Frank E. Curtis, and Jorge Nocedal. Optimization methods for large-scale machine learning. *SIAM Review*, 60(2):223–311, 2018. doi: 10.1137/16M1080173.

[3] Ashok Cutkosky and Francesco Orabona. Momentum-based variance reduction in non-convex SGD. In *Advances in Neural Information Processing Systems*, volume 32, 2019. URL https://arxiv.org/abs/1905.10018.

[4] Aaron Defazio, Francis Bach, and Simon Lacoste-Julien. SAGA: A fast incremental gradient method with support for non-strongly convex composite objectives, 2014. URL https://arxiv.org/abs/1407.0202.

[5] Cong Fang, Chris Junchi Li, Zhouchen Lin, and Tong Zhang. SPIDER: Near-optimal non-convex optimization via stochastic path-integrated differential estimator. In *Advances in Neural Information Processing Systems*, volume 31, 2018. URL https://proceedings.neurips.cc/paper/2018/hash/1543843a4723ed2ab08e18053ae6dc5b-Abstract.html.

[6] Adrien Fradin, Abdurakhmon Sadiev, Laurent Condat, and Peter Richtárik. Tight lower bounds and optimal algorithms for stochastic nonconvex optimization with heavy-tailed noise. In Emtiyaz Khan, Yingzhen Li, Arno Solin, and Aaditya Ramdas, editors, *Proceedings of The 29th International Conference on Artificial Intelligence and Statistics*, volume 300 of *Proceedings of Machine Learning Research*, pages 2548–2556. PMLR, 02–05 May 2026. URL https://proceedings.mlr.press/v300/fradin26a.html.

[7] Saeed Ghadimi and Guanghui Lan. Stochastic first- and zeroth-order methods for nonconvex stochastic programming. *SIAM Journal on Optimization*, 23(4):2341–2368, 2013. doi: 10.1137/120880811.

[8] Eduard Gorbunov, Konstantin P. Burlachenko, Zhize Li, and Peter Richtárik. MARINA: Faster non-convex distributed learning with compression. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 3788–3798. PMLR, 2021. URL https://proceedings.mlr.press/v139/gorbunov21a.html.

[9] Samuel Horváth, Lihua Lei, Peter Richtárik, and Michael I. Jordan. Adaptivity of stochastic gradient methods for nonconvex optimization. *SIAM Journal on Mathematics of Data Science*, 4(2):634–648, 2022. doi: 10.1137/21M1394308.

[10] Rie Johnson and Tong Zhang. Accelerating stochastic gradient descent using predictive variance reduction. In C.J. Burges, L. Bottou, M. Welling, Z. Ghahramani, and K. Weinberger, editors, *Advances in Neural Information Processing Systems*, volume 26. Curran Associates, Inc., 2013. URL https://proceedings.neurips.cc/paper_files/paper/2013/file/ac1dd209cbcc5e5d1c6e28598e8cbbe8-Paper.pdf.

[11] Ahmed Khaled and Chi Jin. Faster federated optimization under second-order similarity. In *International Conference on Learning Representations*, 2023. URL https://openreview.net/forum?id=ElC6LYO4MfD.

[12] Ahmed Khaled and Peter Richtárik. Better theory for SGD in the nonconvex world. *Transactions on Machine Learning Research*, 2023. URL https://openreview.net/pdf?id=AU4qHN2VkS.

[13] Zhize Li, Hongyan Bao, Xiangliang Zhang, and Peter Richtárik. PAGE: A simple and optimal probabilistic gradient estimator for nonconvex optimization. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 6286–6295. PMLR, 2021. URL https://proceedings.mlr.press/v139/li21a.html.

[14] Lam M. Nguyen, Jie Liu, Katya Scheinberg, and Martin Takáč. SARAH: A novel method for machine learning problems using stochastic recursive gradient. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pages 2613–2621. PMLR, 2017. URL https://proceedings.mlr.press/v70/nguyen17b.html.

[15] Lam M. Nguyen, Jie Liu, Katya Scheinberg, and Martin Takáč. Stochastic recursive gradient algorithm for nonconvex optimization, 2017. URL https://arxiv.org/abs/1705.07261.

[16] Lam M. Nguyen, Katya Scheinberg, and Martin Takáč. Inexact SARAH algorithm for stochastic optimization. *Optimization Methods and Software*, 36(1):237–258, 2021. doi: 10.1080/10556788.2020.1818081.

[17] Guannan Qu and Na Li. Harnessing smoothness to accelerate distributed optimization. *IEEE Transactions on Control of Network Systems*, 5(3):1245–1260, September 2018. ISSN 2372-2533. doi: 10.1109/tcns.2017.2698261. URL http://dx.doi.org/10.1109/TCNS.2017.2698261.

[18] Egor Shulgin, Tamaz Gadaev, Sarit Khirirat, and Peter Richtárik. Understanding MARS: When scaling momentum correction provably helps. In Tong Zhang, Miroslav Dudik, Martin Jaggi, Alekh Agarwal, Sharon Li, Dale Schuurmans, Jerry Zhu, Felix Berkenkamp, Hanze Dong, and Alberto Bietti, editors, *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of *Proceedings of Machine Learning Research*, pages 112936–112965. PMLR, 06–11 Jul 2026. URL https://proceedings.mlr.press/v306/shulgin26b.html.

[19] Igor Sokolov. Non-convex stochastic optimization with biased gradient estimators. Master's thesis, King Abdullah University of Science and Technology, 2022.

[20] Rafał Szlendak, Alexander Tyurin, and Peter Richtárik. Permutation compressors for provably faster distributed nonconvex optimization. In *International Conference on Learning Representations*, 2022. URL https://openreview.net/forum?id=GugZ5DzzAu.

[21] Huizhuo Yuan, Yifeng Liu, Shuang Wu, Zhou Xun, and Quanquan Gu. MARS: Unleashing the power of variance reduction for training large models. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 73553–73587. PMLR, 2025. URL https://proceedings.mlr.press/v267/yuan25f.html.

A Detailed Convergence Proofs

A.1 Descent and Weighted Accumulation

Fix deterministic x_0, horizon $K \geq 1$, and constant step $\eta > 0$. Finite second moments follow inductively from smoothness, similarity, and $E_0 < \infty$. Write $q_k = \mathbb{E}[\|d_{k+1}\|^2]$. Conditional

unbiasedness gives $\mathbb{E}[\langle e_k, d_{k+1}\rangle] = 0$, so $E_{k+1} = E_k + q_k$. Smoothness and $2\langle \nabla f(x_k), g_k\rangle = \|\nabla f(x_k)\|^2 + \|g_k\|^2 - \|e_k\|^2$ give (4.5). Summing the error identity yields

$$\sum_{k=0}^{K-1} E_k = KE_0 + \sum_{t=0}^{K-2} (K-1-t)q_t, \tag{A.1}$$

where the sum is empty for $K = 1$, and similarity gives

$$q_t \le \delta^2 \mathbb{E}[\|x_{t+1} - x_t\|^2] = \delta^2\eta^2 M_t. \tag{A.2}$$

A.2 Proof of Theorem 4.1

Sum (4.5), substitute (A.1)–(A.2), and collect the coefficients of M_k. This gives

$$\frac{1}{K}\sum_{k=0}^{K-1} G_k + \frac{2F_K}{\eta K} + \frac{1}{K}\sum_{k=0}^{K-1} c_k M_k \le \frac{2\Delta_0}{\eta K} + E_0, \tag{A.3}$$

where $c_k = 1 - L\eta - \delta^2\eta^2(K-1-k)$. The stepsize condition makes every c_k nonnegative. Dropping these terms and $F_K \ge 0$ proves the theorem.

A.3 Exact Optimized TSGD Guarantee

The largest positive step satisfying $L\eta + (K-1)\delta^2\eta^2 \le 1$ is

$$\boxed{\eta^\star = \frac{2}{L + \sqrt{L^2 + 4(K-1)\delta^2}}.} \tag{A.4}$$

This is the positive quadratic root when $(K-1)\delta^2 > 0$ and equals $1/L$ otherwise. It optimizes the sufficient condition, not the actual stability limit. Substitution into (4.4) yields

$$\overline{G}_K^{\mathrm{TSGD}} \le \frac{\Delta_0}{K}\left(L + \sqrt{L^2 + 4(K-1)\delta^2}\right) + E_0. \tag{A.5}$$

Using

$$\sqrt{L^2 + 4(K-1)\delta^2} \le L + 2\delta\sqrt{K-1}, \qquad \frac{\sqrt{K-1}}{K} \le \frac{1}{\sqrt{K}},$$

gives (4.6).

A.4 SGD under Bounded Variance

Under Assumptions A1–A3, let F_k, G_k now refer to the SGD iterates. Conditional unbiasedness and BV imply

$$\mathbb{E}_\xi[\|h(x_k,\xi)\|^2 \mid x_k] \le \|\nabla f(x_k)\|^2 + \sigma^2.$$

Consequently,

$$F_{k+1} \le F_k - \eta\left(1 - \frac{L\eta}{2}\right) G_k + \frac{L\eta^2\sigma^2}{2}.$$

Summing and dropping $F_K \ge 0$ yields, for $0 < \eta < 2/L$,

$$\boxed{\overline{G}_K^{\mathrm{SGD}} \le \frac{2\Delta_0 + KL\eta^2\sigma^2}{\eta K(2 - L\eta)}.} \tag{A.6}$$

In particular, the fixed gradient-descent step $\eta = 1/L$ gives

$$\overline{G}_K^{\mathrm{SGD}} \le \frac{2L\Delta_0}{K} + \sigma^2. \tag{A.7}$$

For $\Delta_0 > 0$, differentiating the bound shows that its minimum satisfies $L\sigma^2\eta^2 = (2\Delta_0/K)(1-L\eta)$, giving

$$\boxed{\eta^*_{\mathrm{BV},K} = \frac{2}{L + \sqrt{L^2 + 2KL\sigma^2/\Delta_0}}.} \tag{A.8}$$

At this step the bound equals $2\Delta_0/(K\eta)$, hence

$$\boxed{\overline{G}_K^{\mathrm{SGD}} \le \frac{\Delta_0}{K}\left(L + \sqrt{L^2 + \frac{2KL\sigma^2}{\Delta_0}}\right) \le \frac{2L\Delta_0}{K} + \sqrt{\frac{2L\Delta_0\sigma^2}{K}}.} \tag{A.9}$$

The last inequality uses $\sqrt{a+b} \le \sqrt{a} + \sqrt{b}$. For $\sigma = 0$ the step is $1/L$; for $\sigma > 0$ it decreases with K. If $\Delta_0 = 0$, the initial point is already stationary and the optimized formula is not used.

A.5 Iteration Complexity

Fix an accuracy target $\varepsilon > 0$ and let R be uniform on $\{0, \dots, K-1\}$, independent of the run. Then $\mathbb{E}[\|\nabla f(x_R)\|^2] = \overline{G}_K$.

Corollary A.1 (Sufficient TSGD horizon)**.** *Under Theorem 4.1, suppose $E_0 \le \varepsilon^2/3$ and use $\eta = \eta^\star$. An integer horizon satisfying*

$$K \ge \max\left\{1, \frac{6L\Delta_0}{\varepsilon^2}, \frac{36\delta^2\Delta_0^2}{\varepsilon^4}\right\} \tag{A.10}$$

ensures $\mathbb{E}[\|\nabla f(x_R)\|^2] \le \varepsilon^2$.

Proof. Each of the first two terms in (4.6) is at most $\varepsilon^2/3$ under (A.10); the initialization term satisfies the same bound by assumption. □

For SGD with $\Delta_0 > 0$ and step (A.8), the analogous sufficient horizon is

$$K \ge \max\left\{1, \frac{4L\Delta_0}{\varepsilon^2}, \frac{8L\Delta_0\sigma^2}{\varepsilon^4}\right\}. \tag{A.11}$$

Each of the two terms in the simplified SGD bound is then at most $\varepsilon^2/2$. These are iteration upper bounds; initialization and per-iteration oracle costs are accounted for in Appendix E.

B Exact Quadratic Dynamics and Error Sources

B.1 The Quadratic Model

Let x_0 be deterministic, $\eta > 0$ constant, and

$$f(x) = \frac{\lambda}{2}x^2, \qquad h(x; a, b) = (\lambda + a)x + b, \qquad \lambda > 0. \tag{B.1}$$

The noises are independent and satisfy

$$\mathbb{E}[a] = \mathbb{E}[b] = 0, \qquad \mathbb{E}[a^2] = \delta^2, \qquad \mathbb{E}[b^2] = \sigma^2.$$

Thus $\nabla f(x) = \lambda x$ and $\mathbb{E}[h(x; a, b)] = \nabla f(x)$. Each transport step uses a fresh pair (a_{k+1}, b_{k+1}), independent of the past and the square-integrable initial estimate g_0, at both endpoints. The additive noise cancels:

$$D_{k+1} = (\lambda + a_{k+1})(x_{k+1} - x_k). \tag{B.2}$$

Consequently,

$$\mathbb{E}\big[|D_{k+1} - [\nabla f(x_{k+1}) - \nabla f(x_k)]|^2 \mid \mathcal{F}_k\big] = \delta^2 |x_{k+1} - x_k|^2.$$

Assumption A6 holds with equality at $\delta = \sqrt{\mathbb{E}[a^2]}$, independently of σ. Substituting $x_{k+1} - x_k = -\eta g_k$ in $g_{k+1} = g_k + D_{k+1}$ gives

$$g_{k+1} = (1 - \eta\lambda - \eta a_{k+1}) g_k, \qquad e_{k+1} = e_k - \eta a_{k+1} g_k. \tag{B.3}$$

We retain $E_k = \mathbb{E}[|e_k|^2]$, $M_k = \mathbb{E}[|g_k|^2]$, and $G_k = \mathbb{E}[|\nabla f(x_k)|^2]$. In particular, tracking error E_k and true stationarity error G_k are distinct quantities.

B.2 Exact Initialization and Exact Transport

Exact initialization and transport give $g_k = \nabla f(x_k)$ by induction, so TSGD is GD path by path. At $\eta = 1/L$, GD descent yields

$$\frac{1}{2L} \sum_{k=0}^{K-1} \|\nabla f(x_k)\|^2 \le f(x_0) - f(x_K) \le \Delta_0.$$

Consequently,

$$\frac{1}{K} \sum_{k=0}^{K-1} \|\nabla f(x_k)\|^2 \le \frac{2L\Delta_0}{K}. \tag{B.4}$$

The partial sums are uniformly bounded, so the infinite series converges and $\|\nabla f(x_k)\|^2 \to 0$, without convexity or convergence of the iterates.

For the quadratic, $\delta = 0$ means $a = 0$ almost surely. Writing $r = 1 - \eta\lambda$, exact initialization gives

$$x_k = r^k x_0, \qquad g_k = \lambda r^k x_0, \qquad E_k = 0. \tag{B.5}$$

Thus $G_k = \lambda^2 r^{2k} x_0^2 \to 0$ for $0 < \eta < 2/\lambda$.

B.3 Inexact Initialization and Exact Transport

For any objective, let $g_0 = \nabla f(x_0) + e_0$ and take exact differences. Then $e_k = e_0$ for every k, and

$$g_k = \nabla f(x_k) + e_0 = \nabla \widetilde{f}_{e_0}(x_k), \qquad \widetilde{f}_{e_0}(x) := f(x) + \langle e_0, x \rangle.$$

Hence TSGD is GD on a fixed linearly tilted objective, and $E_k = E_0$. At $\eta = 1/L$, Theorem 4.1 gives

$$\frac{1}{K} \sum_{k=0}^{K-1} G_k \le \frac{2L\Delta_0}{K} + E_0. \tag{B.6}$$

For fixed e_0, if $\widetilde{f}_{e_0}$ is also bounded below, GD descent on that objective implies $\nabla f(x_k) \to -e_0$ and $\|\nabla f(x_k)\|^2 \to \|e_0\|^2$. This additional lower bound is required for the pathwise asymptotic claim, not for (B.6).

For the quadratic and $\delta = 0$, the exact dynamics are

$$g_k = r^k(\lambda x_0 + e_0), \qquad x_k = r^k x_0 - \frac{1 - r^k}{\lambda} e_0. \tag{B.7}$$

The first identity follows from (B.3); the second follows from $\lambda x_k = g_k - e_0$. If $\mathbb{E}[e_0] = 0$, then

$$G_k = r^{2k}\lambda^2 x_0^2 + (1 - r^k)^2 E_0. \tag{B.8}$$

For $0 < \eta < 2/\lambda$, even with biased square-integrable e_0, $x_k \to -e_0/\lambda$ in mean square and

$$\lim_{k\to\infty} G_k = E_0. \tag{B.9}$$

Sharpness of the initialization coefficient. Choose an initialization with $E_0 > 0$. For $\varepsilon > 0$, choose N with $|G_k - E_0| \le \varepsilon$ for $k \ge N$. For every $K > N$,

$$\left| \frac{1}{K} \sum_{k=0}^{K-1} G_k - E_0 \right| \le \frac{1}{K} \sum_{k=0}^{N-1} |G_k - E_0| + \varepsilon.$$

Letting $K \to \infty$ and then $\varepsilon \downarrow 0$ gives, at the admissible $\eta = 1/L = 1/\lambda$,

$$\lim_{K\to\infty} \frac{1}{K} \sum_{k=0}^{K-1} G_k = E_0 = \lim_{K\to\infty} \left(\frac{2\Delta_0}{\eta K} + E_0 \right).$$

Thus the coefficient one on E_0 in (4.4) is sharp.

B.4 Exact Initialization and Stochastic Transport

Take $g_0 = \lambda x_0$ and $\delta > 0$, and define

$$r := 1 - \eta\lambda, \qquad \rho := \mathbb{E}[(r - \eta a)^2] = r^2 + \eta^2\delta^2 = 1 - 2\eta\lambda + \eta^2(\lambda^2 + \delta^2).$$

Independence and centeredness of a_{k+1} give

$$M_{k+1} = \rho M_k, \qquad E_{k+1} = E_k + \eta^2\delta^2 M_k. \tag{B.10}$$

Starting from $E_0 = 0$ and $M_0 = \lambda^2 x_0^2$,

$$M_k = \rho^k \lambda^2 x_0^2, \qquad E_k = \eta^2\delta^2\lambda^2 x_0^2 \sum_{t=0}^{k-1} \rho^t. \tag{B.11}$$

The stability condition is

$$\rho < 1 \quad \Longleftrightarrow \quad 0 < \eta < \frac{2\lambda}{\lambda^2 + \delta^2}. \tag{B.12}$$

Under this condition, $g_k \to 0$ in mean square, while

$$E_\infty := \lim_{k\to\infty} E_k = \frac{\eta\delta^2\lambda^2 x_0^2}{2\lambda - \eta(\lambda^2 + \delta^2)}. \tag{B.13}$$

To relate this tracking error to the true gradient, note that $e_{t+1} - e_t = -\eta a_{t+1} g_t$ are orthogonal martingale increments. For $m > n$,

$$\mathbb{E}[|e_m - e_n|^2] = \eta^2\delta^2 \sum_{t=n}^{m-1} M_t \le \frac{\eta^2\delta^2 M_0}{1 - \rho} \rho^n \longrightarrow 0.$$

Thus (e_k) is Cauchy in the complete space L^2, and converges in mean square to some e_∞. Since $g_k \to 0$ in mean square, $x_k = (g_k - e_k)/\lambda \to -e_\infty/\lambda =: x_\infty$ in mean square as well. Consequently,

$$\lim_{k\to\infty} M_k = 0, \qquad \lim_{k\to\infty} G_k = \mathbb{E}[|\nabla f(x_\infty)|^2] = \mathbb{E}[|e_\infty|^2] = E_\infty > 0 \quad (x_0 \neq 0).$$

The estimator converges to zero in mean square while the true gradient does not. This is the false-convergence phenomenon.

B.5 Inexact Initialization and Stochastic Transport

Allow arbitrary square-integrable e_0 and retain (B.12). The moment recursions remain valid. Define

$$A := \frac{\eta^2\delta^2}{1-\rho} = \frac{\eta\delta^2}{2\lambda - \eta(\lambda^2+\delta^2)}. \tag{B.14}$$

Since $A(1-\rho) = \eta^2\delta^2$,

$$E_{k+1} + AM_{k+1} = E_k + (\eta^2\delta^2 + A\rho)M_k = E_k + AM_k.$$

Induction proves the conservation law

$$E_k + AM_k = E_0 + AM_0 \qquad (k \geq 0). \tag{B.15}$$

Since $M_k = \rho^k M_0 \to 0$, the exact interaction formula follows:

$$\boxed{E_\infty = E_0 + \frac{\eta\delta^2}{2\lambda - \eta(\lambda^2+\delta^2)} M_0 = E_0 + AM_0.} \tag{B.16}$$

The preceding L^2 argument also gives $G_k \to E_\infty$. Initialization contributes directly through E_0 and indirectly through the initial estimator magnitude

$$M_0 = \lambda^2 x_0^2 + 2\lambda x_0 \mathbb{E}[e_0] + E_0.$$

Bias can therefore increase or decrease the transport contribution. For $g_0 = \lambda x_0 + b_0$, with centered b_0 independent of future samples and $\mathbb{E}[b_0^2] = \sigma^2$, the three contributions are

$$E_\infty = \underbrace{\sigma^2}_{\text{initialization}} + \underbrace{A\lambda^2 x_0^2}_{\text{transport}} + \underbrace{A\sigma^2}_{\text{interaction}}. \tag{B.17}$$

In Table 1, the four corresponding limiting stationarity errors are respectively 0, $A\lambda^2 x_0^2$, E_0, and $E_0 + AM_0$. The exact-transport identities are general; these stochastic-transport formulas rely on the quadratic model.

B.6 Initialization and Transport Errors on Least Squares

Figure 3 separates the effects of initialization and similarity in the least-squares model of Appendix D.6. We prescribe $g_0 = Hx_0 + e_0$, with $e_0 \sim \mathcal{N}(0, (E_0/d)I_d)$ independent of subsequent samples. Each coordinate follows the scalar quadratic recursion with curvature λ_j and noise variance $\kappa^2\lambda_j^2$, so summing (B.16) gives

$$\lim_{k\to\infty} \mathbb{E}[\|\nabla f(x_k)\|^2] = E_0 + \sum_{j=1}^{d} \frac{\eta\kappa^2\lambda_j}{2 - \eta(1+\kappa^2)\lambda_j} \left(\lambda_j^2 x_{0,j}^2 + \frac{E_0}{d}\right), \tag{B.18}$$

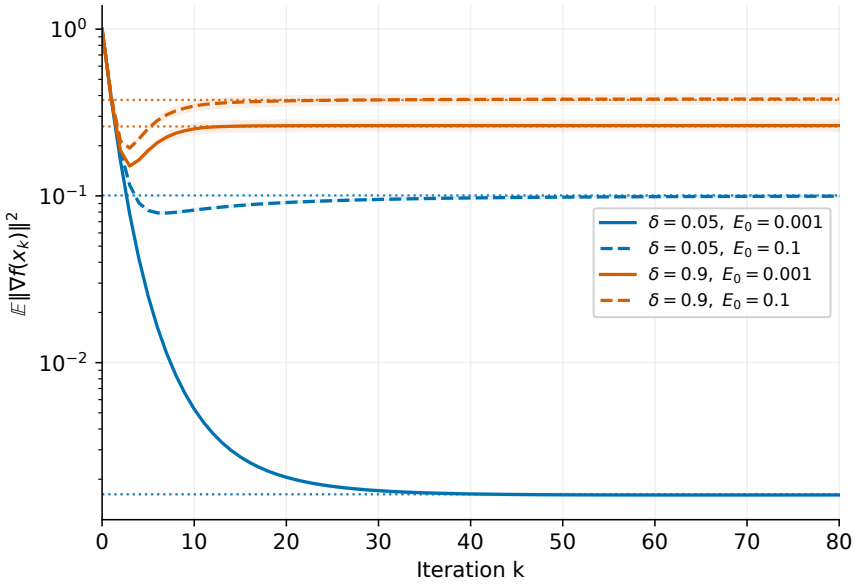


Figure 3: Initialization and transport errors on vector least squares. Each curve uses one of $\delta \in \{0.05, 0.9\}$ and $E_0 \in \{10^{-3}, 10^{-1}\}$; dotted lines are the exact limiting stationarity errors in (B.18). Means over 4,000 trajectories with pointwise bands of ± 2 standard errors.

provided $0 < \eta < 2/[(1+\kappa^2)\max_j \lambda_j]$, as in our runs.

We compare $\delta = \kappa \in \{0.05, 0.9\}$ and $E_0 \in \{10^{-3}, 10^{-1}\}$. The observed plateaus agree with the exact limits shown by dotted lines. Better initialization reduces both the inherited error and its contribution to later transport error; smaller δ reduces the transport contribution. Neither improvement replaces the other: accurate initialization alone does not prevent a large transport residual. These controlled initializations isolate the two effects without modeling the cost of obtaining g_0.

C Noise Assumptions

C.1 Bounded Variance, ABC, and Centered Similarity

Write $h(x, \xi) = \nabla f(x) + r(x, \xi)$ with $\mathbb{E}_\xi[r(x, \xi)] = 0$.

ABC generalizes BV. BV requires the gradient-noise variance to be uniformly bounded over the domain. The more general ABC condition, introduced as expected smoothness by Khaled and Richtárik [12], assumes that some $A, B, C \geq 0$ satisfy

$$\mathbb{E}_\xi[\|h(x, \xi)\|^2] \leq 2A(f(x) - f_{\inf}) + B\|\nabla f(x)\|^2 + C \qquad \text{for all } x. \tag{C.1}$$

Unbiasedness gives

$$\mathbb{E}_\xi[\|h(x, \xi)\|^2] = \|\nabla f(x)\|^2 + \mathbb{E}_\xi[\|r(x, \xi)\|^2].$$

BV is the special case $(A, B, C) = (0, 1, \sigma^2)$. ABC permits $\mathbb{E}_\xi[\|h(x, \xi)\|^2]$ to grow with $f(x) - f_{\inf}$ or $\|\nabla f(x)\|^2$. In particular, if $C = 0$ and a sequence (x_k) satisfies

$$f(x_k) - f_{\inf} \to 0 \qquad \text{and} \qquad \|\nabla f(x_k)\|^2 \to 0,$$

then ABC implies

$$\mathbb{E}_\xi[\|h(x_k,\xi)\|^2] \to 0.$$

Thus the second-moment term in SGD's descent bound can vanish instead of being controlled by a fixed noise bound.

For example, let $f(x) = x^2/2$ and $h(x,a) = ax$, where a equals 0 or 2 with equal probability. Its residual is $r(x,a) = h(x,a) - \nabla f(x) = (a-1)x$, so

$$\mathbb{E}[h(x,a)] = x, \qquad \mathbb{E}[|h(x,a)|^2] = 2x^2, \qquad \mathbb{E}[|r(x,a)|^2] = x^2.$$

Thus ABC holds with $(A,B,C) = (0,2,0)$, while global BV fails because $\mathbb{E}[|h(x,a) - \nabla f(x)|^2] = \mathbb{E}[|r(x,a)|^2] = x^2$ is unbounded as $|x| \to \infty$. As in Section 3.3, fresh gradients become exact at the minimizer despite unbounded noise variance over the domain.

One-point and two-point conditions. BV and ABC constrain the stochastic gradient at each point separately. Centered similarity instead constrains noise differences under shared sampling, and therefore depends on how the oracle couples two points. In Section 3.2, the pointwise gradient distributions are unchanged as the frequency ω varies, while the smallest similarity constant is $\delta = \sigma\omega$. Thus the same pointwise distributions can coexist with arbitrarily different transport accuracy.

The counterexamples below concern global oracle conditions and show that neither BV nor ABC implies centered similarity, while centered similarity implies neither BV nor ABC.

Neither BV nor ABC implies global similarity. Take $f(x) = x^2/2$ and $h(x,\xi) = x + \xi\sin(x^2)$, where $\mathbb{P}(\xi = -1) = \mathbb{P}(\xi = 1) = 1/2$. The oracle is unbiased and

$$\mathbb{E}_\xi[|h(x,\xi) - \nabla f(x)|^2] = \sin^2(x^2) \le 1,$$

so BV holds with $\sigma^2 = 1$, and ABC holds with $(A,B,C) = (0,1,1)$. Under shared sampling,

$$\mathbb{E}_\xi[|r(x,\xi) - r(y,\xi)|^2] = \big(\sin(x^2) - \sin(y^2)\big)^2.$$

A finite similarity constant would imply

$$\begin{aligned} |\sin(x^2) - \sin(y^2)| &\le \delta|x-y| && \text{for every } x, y, \\ |2x\cos(x^2)| = \lim_{y\to x} \frac{|\sin(y^2) - \sin(x^2)|}{|y-x|} &\le \delta && \text{for every } x. \end{aligned}$$

However, at $x_n = \sqrt{2\pi n}$, $n \ge 1$, this would require

$$|2x_n\cos(x_n^2)| = 2\sqrt{2\pi n} \le \delta \qquad \text{for every } n.$$

For any finite δ, choosing an integer $n > \delta^2/(8\pi)$ violates this inequality. Hence no finite global similarity constant exists.

Global similarity implies neither BV nor ABC. Take $f(x) = \log\cosh x$ and $h(x,\xi) = \tanh x + \xi x$, with $\mathbb{E}[\xi] = 0$ and $\mathbb{E}[\xi^2] = \delta^2 > 0$. The objective is lower bounded and 1-smooth, since

$$f(x) \ge 0, \qquad \nabla f(x) = \tanh x, \qquad f''(x) = \frac{1}{\cosh^2 x} \in (0,1].$$

The unbiased oracle has residual $r(x,\xi) = \xi x$, and hence

$$\mathbb{E}_\xi[|r(x,\xi) - r(y,\xi)|^2] = \delta^2(x-y)^2, \qquad \mathbb{E}_\xi[|r(x,\xi)|^2] = \delta^2 x^2.$$

Similarity holds with equality, but no finite global BV constant exists. Since $f_{\inf} = 0$, ABC would require

$$\tanh^2 x + \delta^2 x^2 \le 2A \log\cosh x + B\tanh^2 x + C \le 2A|x| + B + C \qquad \text{for all } x.$$

For $x \neq 0$, dividing by x^2 and dropping the nonnegative term $\tanh^2 x/x^2$ gives

$$\delta^2 \le \frac{2A}{|x|} + \frac{B+C}{x^2}.$$

For any finite nonnegative A, B, C, the right-hand side tends to zero as $|x| \to \infty$, whereas the left-hand side is the fixed positive constant δ^2. The inequality therefore fails for sufficiently large $|x|$, so no finite ABC constants exist.

For this oracle, Theorem 4.1 still applies: the coupled increment has conditional error variance $\delta^2(x_{k+1} - x_k)^2$. Thus exact initialization gives the $O(K^{-1/2})$ TSGD guarantee despite failure of both BV and ABC. This does not by itself establish that SGD fails.

C.2 Quadratic Finite Sums and the Smallest Similarity Constant

For $f_i(x) = \frac{1}{2}x^\top A_i x + b_i^\top x + c_i$ with symmetric A_i, let $f = \frac{1}{n}\sum_{i=1}^n f_i$, $\bar{A} = \frac{1}{n}\sum_{i=1}^n A_i$, and sample i uniformly. The centered gradient difference is

$$[\nabla f_i(x) - \nabla f_i(y)] - [\nabla f(x) - \nabla f(y)] = (A_i - \bar{A})(x-y). \tag{C.2}$$

For $z = x - y$,

$$\mathbb{E}_i[\|(A_i - \bar{A})z\|^2] = z^\top \left(\frac{1}{n}\sum_{i=1}^n (A_i - \bar{A})^2\right) z.$$

Taking the supremum over $\|z\| = 1$ gives the smallest admissible global constant,

$$\delta_{\min}^2 = \lambda_{\max}\left(\frac{1}{n}\sum_{i=1}^n (A_i - \bar{A})^2\right). \tag{C.3}$$

The linear terms cancel irrespective of their magnitude, while the smoothness constant of the average quadratic is $L = \|\bar{A}\|_{\mathrm{op}}$. There is no fixed ordering between δ and L. Identical A_i give $\delta_{\min} = 0$ even with large L and arbitrarily different b_i. Conversely, for $n \ge 3$ scalar components, take $f_1(x) = nLx^2/2$ and $f_i(x) = 0$ for $i \ge 2$. Then $f(x) = Lx^2/2$ but

$$\delta_{\min}^2 = \frac{1}{n}((nL - L)^2 + (n-1)L^2) = (n-1)L^2 > L^2.$$

Thus similarity measures the discrepancy between stochastic and deterministic gradient changes, rather than the smoothness of their average.

D Detailed Illustrative Examples

D.1 Additive Noise: Exact Trajectories and Horizon Tuning

For explicit constants, specialize the additive example to

$$f(x) = \frac{x^2}{2}, \qquad h(x,\xi) = x + \xi, \qquad \mathbb{P}(\xi = 1) = \mathbb{P}(\xi = -1) = \frac{1}{2}, \qquad x_0 = 1. \tag{D.1}$$

Here $L = 1$, $\Delta_0 = 1/2$, $\sigma^2 = 1$, and $\delta = 0$. At $\eta = 1$, SGD satisfies $x_{k+1} = -\xi_k$, hence

$$G_k^{\mathrm{SGD}} = 1 \quad (k \geq 0), \qquad \boxed{\overline{G}_K^{\mathrm{SGD}} = 1.} \tag{D.2}$$

Thus the fixed-step noise floor is present in the actual trajectory.

For the horizon-dependent BV stepsize (A.8), write $\eta_K = 2/(1 + \sqrt{1 + 4K})$ and $r_K = 1 - \eta_K$. Independence and centeredness of the fresh sample give

$$G_{k+1}^{\mathrm{SGD}} = \mathbb{E}[(r_K x_k - \eta_K \xi_k)^2] = r_K^2 G_k^{\mathrm{SGD}} + \eta_K^2.$$

Solving from $G_0 = 1$ and averaging the geometric terms yields

$$\begin{aligned} G_k^{\mathrm{SGD}} &= r_K^{2k} + \frac{\eta_K}{2 - \eta_K}(1 - r_K^{2k}), \\ \overline{G}_K^{\mathrm{SGD}} &= \frac{\eta_K}{2 - \eta_K} + \left(1 - \frac{\eta_K}{2 - \eta_K}\right) \frac{1 - r_K^{2K}}{K(1 - r_K^2)}. \end{aligned}$$

Since $\eta_K \sim K^{-1/2}$, $r_K^{2K} \leq e^{-2K\eta_K} \to 0$ and $1 - r_K^2 = \eta_K(2 - \eta_K) \sim 2K^{-1/2}$. Each term in the average is therefore asymptotic to $1/(2\sqrt{K})$:

$$\boxed{\overline{G}_K^{\mathrm{SGD}} = \frac{1 + o(1)}{\sqrt{K}}.} \tag{D.3}$$

The optimized BV bound is $(1 + \sqrt{1 + 4K})/(2K)$, so both its rate and leading constant are attained by this prescribed run.

For TSGD, use $\eta = 1$ and any square-integrable $g_0 = 1 + e_0$. Exact differences give $D_{k+1} = x_{k+1} - x_k = -g_k$; therefore $g_1 = 0$ and $x_k = -e_0$ for all $k \geq 1$. Hence

$$\boxed{\overline{G}_K^{\mathrm{TSGD}} = \frac{1}{K} + \left(1 - \frac{1}{K}\right) E_0.} \tag{D.4}$$

With exact initialization, TSGD reaches the minimizer after one step. For fixed $E_0 > 0$, horizon-tuned SGD eventually has the smaller error. The comparison depends on initialization as well as transport.

D.2 Multiplicative Noise: Fresh Gradients and False Convergence

For (3.3), $\mathbb{E}[a] = 1$ and $\mathbb{E}[(a-1)^2] = 1$. The oracle is unbiased and its pointwise noise variance is x^2, so global BV does not hold. Coupled centered differences nevertheless satisfy

$$\mathbb{E}[|(a-1)(x-y)|^2] = |x-y|^2,$$

giving $\delta = 1$. These comparisons use exact dynamics rather than the BV theorem.

At $\eta = 1/2$, SGD obeys $x_{k+1} = (1 - a_k/2)x_k$; it remains at one until the first sample $a = 2$, then stays at zero. Thus

$$\boxed{G_k^{\mathrm{SGD}} = 2^{-k}, \qquad \overline{G}_K^{\mathrm{SGD}} = \frac{2(1 - 2^{-K})}{K}.} \tag{D.5}$$

For TSGD at the same step, let $g_0 = \nabla f(x_0) = 1$ and $N := \min\{n \geq 1 : a_n = 2\}$. The estimator recursion is $g_{k+1} = (1 - a_{k+1}/2)g_k$, so

$$\mathbb{P}(N = n) = 2^{-n}, \qquad g_k = \mathbf{1}_{\{k<N\}}, \qquad x_k = 1 - \frac{1}{2}\min\{k, N\}.$$

Since $N < \infty$ almost surely and $\mathbb{E}[N^2] = 6$, dominated convergence gives $x_k \to 1 - N/2$ in mean square. Using $\mathbb{E}[N] = 2$,

$$\boxed{\lim_{k\to\infty} G_k^{\mathrm{TSGD}} = \mathbb{E}[(1 - N/2)^2] = \frac{1}{2}.} \tag{D.6}$$

The estimator becomes zero almost surely in finite time, but the limiting true gradient is nonzero with probability $1 - \mathbb{P}(N = 2) = 3/4$. Cesàro averaging also gives $\overline{G}_K^{\mathrm{TSGD}} \to 1/2$.

D.3 Sharpness for Horizon-Tuned TSGD

For the same oracle with exact initialization, (A.4) gives $\eta_K^\star = 2/(1 + \sqrt{4K - 3})$, since $L = \delta = 1$ and $\Delta_0 = 1/2$. The optimized guarantee becomes

$$\overline{G}_K^{\mathrm{TSGD}} \leq \frac{1 + \sqrt{4K - 3}}{2K}. \tag{D.7}$$

Lemma D.1 (Asymptotic sharpness for tuned TSGD)**.** *For (3.3), with exact initialization and the constant step $\eta_K^\star$ for the K-iteration run,*

$$\lim_{K\to\infty} \sqrt{K}\, \overline{G}_K^{\mathrm{TSGD}} = 1.$$

Proof. Fix $K \geq 2$ and write $\eta = \eta_K^\star$. The updates imply

$$g_{k+1} = (1 - \eta a_{k+1})g_k, \qquad e_{k+1} = e_k + \eta(1 - a_{k+1})g_k.$$

Set $C_k := \mathbb{E}[g_k e_k]$ and $q := \mathbb{E}[(1 - \eta a)^2] = 1 - 2\eta + 2\eta^2$. Independence yields

$$M_{k+1} = qM_k, \qquad E_{k+1} = E_k + \eta^2 M_k, \qquad C_{k+1} = (1 - \eta)C_k + \eta^2 M_k.$$

Starting from $M_0 = 1$ and $E_0 = C_0 = 0$, these give

$$M_k = q^k, \qquad E_k = \frac{\eta}{2(1 - \eta)}(1 - q^k), \qquad C_k = \eta^2 \sum_{t=0}^{k-1} (1 - \eta)^{k-1-t} q^t.$$

Since $0 < \eta < 1$ and $q < 1$, all summands are nonnegative and

$$\sum_{k<K} C_k \leq \eta^2 \left(\sum_{t\geq 0} q^t\right)\left(\sum_{j\geq 0} (1 - \eta)^j\right) = \frac{\eta^2}{(1 - q)\eta} = \frac{1}{2(1 - \eta)}.$$

Thus the average cross moment is $O(K^{-1})$ as $\eta_K^\star \to 0$. For the other moments,

$$\frac{1}{K}\sum_{k<K} M_k = \frac{1-q^K}{K(1-q)}, \qquad \frac{1}{K}\sum_{k<K} E_k = \frac{\eta}{2(1-\eta)}\left(1 - \frac{1}{K}\sum_{k<K} M_k\right).$$

Now $\eta_K^\star \sim K^{-1/2}$, $1-q = 2\eta_K^\star(1-\eta_K^\star) \sim 2K^{-1/2}$, and $q^K \le e^{-K(1-q)} \to 0$. Therefore

$$\frac{1}{K}\sum_{k<K} M_k \sim \frac{1}{2\sqrt{K}}, \qquad \frac{1}{K}\sum_{k<K} E_k \sim \frac{1}{2\sqrt{K}}.$$

Finally, $x_k = g_k - e_k$ implies $G_k = M_k + E_k - 2C_k$. The averaged cross term is lower order, which proves the claim. □

The theorem's bound has the correct leading rate and constant for this prescribed TSGD run. Fixed-step SGD in (D.5) has average error $\Theta(K^{-1})$, whereas this horizon-tuned TSGD run has $\Theta(K^{-1/2})$ error. This is not a lower bound over all TSGD stepsize schedules.

D.4 Fixed Variance and Variable Similarity

Lemma D.2 (Fixed variance and variable similarity)**.** *For (3.1), the noise is centered, satisfies (3.2), and has smallest global similarity constant $\delta = \sigma\omega$ for $\omega \ge 0$.*

Proof. A uniform phase makes the cosine have mean zero and second moment $1/2$, so $\mathbb{E}[r(x,\varphi)^2] = 2\sigma^2(1/2) = \sigma^2$. For the coupled difference,

$$r(x,\varphi) - r(y,\varphi) = -2\sqrt{2}\sigma \sin\left(\frac{\omega(x+y)}{2} + \varphi\right)\sin\left(\frac{\omega(x-y)}{2}\right).$$

Squaring and using $\mathbb{E}_\varphi[\sin^2(c+\varphi)] = 1/2$ for every fixed c gives

$$\mathbb{E}[|r(x,\varphi) - r(y,\varphi)|^2] = 8\sigma^2 \cdot \frac{1}{2}\sin^2\left(\frac{\omega(x-y)}{2}\right) = 4\sigma^2 \sin^2\left(\frac{\omega(x-y)}{2}\right).$$

This verifies (3.2). The inequality $|\sin t| \le |t|$ proves similarity with $\delta = \sigma\omega$; minimality follows from

$$\lim_{y\to x} \frac{4\sigma^2 \sin^2(\omega(x-y)/2)}{(x-y)^2} = \sigma^2\omega^2.$$

□

SGD's fresh noise has conditional variance σ^2 at every iterate, giving the frequency-independent exact recursion

$$G_{k+1}^{\mathrm{SGD}} = (1-L\eta)^2 G_k^{\mathrm{SGD}} + L^2\eta^2\sigma^2. \tag{D.8}$$

Indeed, even its conditional noise distribution is independent of x_k and ω, since adding ωx_k to a uniform phase preserves its distribution modulo 2π. For TSGD,

$$\begin{aligned} E_{k+1} - E_k &= 4\sigma^2\mathbb{E}\left[\sin^2\left(\frac{\omega(x_{k+1}-x_k)}{2}\right)\right] \\ &\le \sigma^2\omega^2\eta^2 M_k. \end{aligned} \tag{D.9}$$

For fixed $\sigma > 0$, varying ω realizes any smallest global similarity constant without changing the pointwise noise distribution. The sine factor also shows that increasing ω need not increase every realized increment's error monotonically.

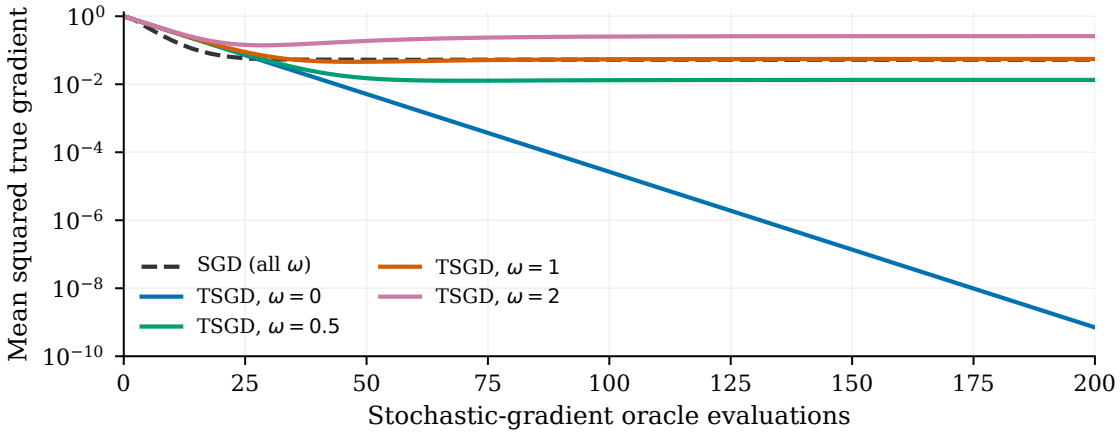


Figure 4: Fixed pointwise noise, variable similarity. Changing ω leaves SGD's law unchanged but changes TSGD's transport accuracy. All methods use exact antithetic initialization and $\eta = 0.1$; the horizontal axis includes initialization costs.

D.5 Numerical Illustration of Fixed Variance and Variable Similarity

Can noise variation change the relative performance of SGD and TSGD when pointwise noise stays the same? We use the oracle (3.1) with $L = \sigma = x_0 = 1$, $\eta = 0.1$, and $\omega \in \{0, 0.5, 1, 2\}$, so $\delta = \omega$. All methods start from the exact estimate

$$g_0 = \tfrac{1}{2}(h(x_0, \varphi_0) + h(x_0, \varphi_0 + \pi)) = Lx_0.$$

This initialization exploits the oracle's antithetic symmetry; two independent samples would not give an exact gradient. Subsequent phases are independent and uniform, with a shared phase within each transport difference. The horizontal axis counts the two initialization evaluations, then one evaluation per subsequent SGD update and two per TSGD update. Curves average 10^5 independent runs, with pointwise bands of ± 2 Monte Carlo standard errors.

Figure 4 shows that SGD is unaffected by ω, whereas TSGD follows GD at $\omega = 0$ and accumulates error at nonzero frequencies. At the final budget, TSGD has smaller error than SGD for $\omega = 0.5$ and larger error for $\omega = 2$. Thus identical pointwise noise distributions can conceal very different transport accuracy. This is a constant-stepsize illustration, not a comparison of optimally tuned methods.

D.6 Least-Squares Experiments

We use one least-squares model to illustrate initialization quality, label-noise cancellation, false convergence, and the effect of similarity. The figures compare iterations at a common stepsize, not computational cost or optimally tuned performance.

Common setup. Let $f(x) = \frac{1}{2}\mathbb{E}[\|Ax - b\|^2]$, with

$$H = \mathrm{diag}(0.1, 0.2, \ldots, 1), \qquad A = \sqrt{v}\, H^{1/2}, \qquad b = s\varepsilon,$$

where $v = 1 + \kappa\zeta$, $\mathbb{P}(\zeta = 1) = \mathbb{P}(\zeta = -1) = 1/2$, and $\varepsilon \sim \mathcal{N}(0, I_{10})$ is independent of ζ. The minimizer is $x^\star = 0$, and

$$\nabla f(x) = Hx, \qquad h(x; v, \varepsilon) = vHx - s\sqrt{v}\, H^{1/2}\varepsilon, \qquad h(x; v, \varepsilon) - h(y; v, \varepsilon) = vH(x - y).$$

Thus label noise, controlled by s, cancels from shared-sample differences, while feature variability determines $\delta = \kappa$. We use $x_0 = \mathbf{1}/\|H\mathbf{1}\|$, $\eta = 0.5$, and 4,000 independent trajectories. The figures show the first 80 of 400 updates, with pointwise bands of ± 2 Monte Carlo standard errors. Initial estimates are independent of subsequent samples.

Initialization quality. In Figure 5, $s = 1$ and $\kappa = 0.05$: gradients are noisy but their differences are accurate. TSGD initializes by averaging $B_0 \in \{1, 128, 512\}$ actual stochastic gradients; SGD uses one fresh sample at every update. Single-sample initialization leaves TSGD with a larger residual than SGD, whereas the larger initial batches give smaller residuals. Accurate transport preserves useful initial information but does not correct a poor initial estimate. The extra cost of larger initial batches is not reflected in the iteration axis.

False convergence. Figure 2(b) uses noiseless labels ($s = 0$), $\kappa = 0.9$, and $B_0 = 128$ sampled gradients to initialize TSGD. SGD's true gradient and TSGD's estimator both tend to zero in mean square, but TSGD's true-gradient error remains positive. The plot therefore distinguishes a small estimator from actual stationarity.

Label-noise cancellation. In Figure 2(a), both methods instead start from the exact gradient. We fix $\kappa = 0.2$ and use matched feature draws across $s \in \{0, 1, 5\}$. TSGD's trajectories coincide because label noise cancels from every difference; its remaining error comes from feature variability. SGD's residual grows with label noise and vanishes at $s = 0$.

Initialization and similarity. Figure 3 varies $\delta \in \{0.05, 0.9\}$ and a prescribed initial error $E_0 \in \{10^{-3}, 10^{-1}\}$. Appendix B.6 gives the exact residuals. Unlike the sampled initializations in the first two experiments, this experiment and the label-noise experiment use controlled estimates to isolate mechanisms, without modeling how that accuracy is obtained.

E Oracle Complexity and Additional Algorithmic Details

E.1 Initialization and Accuracy Targets

Suppose an unbiased oracle at x_0 has variance at most σ_0^2. With B_0 independent samples, independent also of subsequent transport samples, set

$$g_0 = \frac{1}{B_0} \sum_{i=1}^{B_0} h(x_0, \xi_i^{(0)}).$$

Independence and centering cancel the cross terms, giving

$$E_0 = \mathbb{E}[\|g_0 - \nabla f(x_0)\|^2] \le \frac{\sigma_0^2}{B_0}.$$

Thus $B_0 \ge \max\{1, \lceil 3\sigma_0^2/\varepsilon^2 \rceil\}$ ensures $E_0 \le \varepsilon^2/3$, as required by Corollary A.1. Only the variance at x_0 is needed for this construction. The main theorem itself requires neither unbiased initialization nor a pointwise oracle.

Count K iterate updates. Single-sample SGD uses K pointwise gradient evaluations, whereas the shared-sample TSGD construction uses at most $B_0 + 2(K-1)$. Its final correction D_K is unused: g_{K-1} suffices to compute x_K, and the output is selected from $x_0, \ldots, x_{K-1}$. Under the common BV

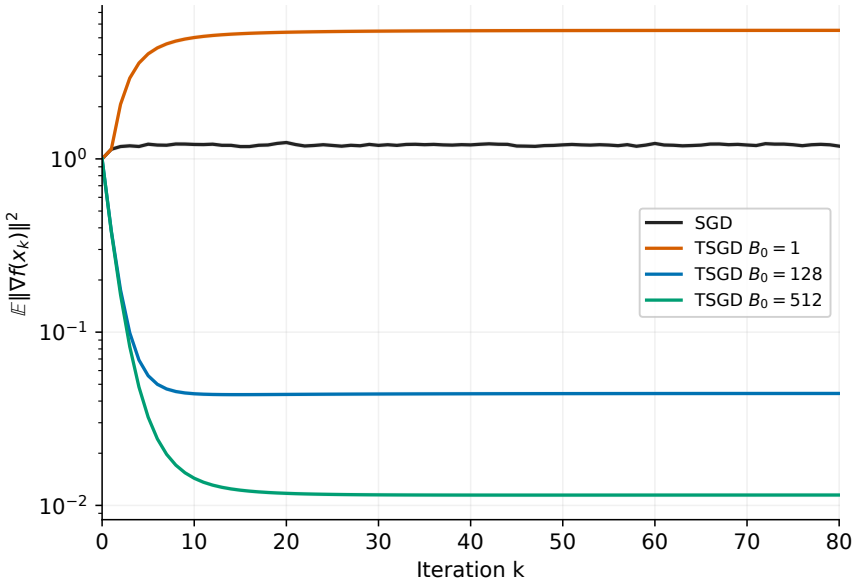


Figure 5: Initialization quality on vector least squares with noisy labels. TSGD is initialized by averaging B_0 independent samples. Means over 4,000 trajectories with pointwise bands of ± 2 standard errors.

assumption, one can take $\sigma_0^2 \le \sigma^2$. Combining the batch size above with (A.10) and (A.11) gives sufficient evaluation budgets of order

$$\begin{aligned} N_{\mathrm{SGD}}(\varepsilon) &= O\left(1 + \frac{L\Delta_0}{\varepsilon^2} + \frac{L\Delta_0\sigma^2}{\varepsilon^4}\right), \\ N_{\mathrm{TSGD}}(\varepsilon) &= O\left(1 + \frac{\sigma^2}{\varepsilon^2} + \frac{L\Delta_0}{\varepsilon^2} + \frac{\delta^2\Delta_0^2}{\varepsilon^4}\right). \end{aligned} \tag{E.1}$$

These are upper bounds for the stated implementations, not oracle lower bounds. The initialization term depends on this choice of g_0 and can change when other initial gradient information is available.

E.2 Comparisons at Equal Iterations and Oracle Budgets

When BV and global similarity both hold, let $U_{\mathrm{SGD}}(K)$ and $U_{\mathrm{TSGD}}(K)$ denote the optimized upper bounds in (A.9) and (A.5). At equal iteration counts, subtracting gives

$$U_{\mathrm{TSGD}}(K) < U_{\mathrm{SGD}}(K) \quad \Longleftrightarrow \quad E_0 < \frac{\Delta_0}{K}\left[\sqrt{L^2 + \frac{2KL\sigma^2}{\Delta_0}} - \sqrt{L^2 + 4(K-1)\delta^2}\right]. \tag{E.2}$$

With exact initialization, this reduces to $2(K-1)\delta^2\Delta_0 < KL\sigma^2$. At an evaluation budget $N \ge \max\{1, B_0\}$, the horizons are instead

$$K_{\mathrm{SGD}} = N, \qquad K_{\mathrm{TSGD}} = 1 + \left\lfloor \frac{N - B_0}{2} \right\rfloor. \tag{E.3}$$

For positive δ, σ, Δ_0, if $B_0 = o(N)$ and $E_0 = o(N^{-1/2})$, the leading bounds are $\sqrt{2L\Delta_0\sigma^2/N}$ for SGD and $2\sqrt{2}\delta\Delta_0/\sqrt{N}$ for TSGD. Thus the leading TSGD bound is smaller when $4\delta^2\Delta_0 < L\sigma^2$,

versus $2\delta^2\Delta_0 < L\sigma^2$ at equal iterations. These are comparisons of upper bounds, not predictions of actual performance. A fixed positive E_0 does not satisfy the asymptotic condition, and exact initialization must also have its cost counted.

E.3 Refreshing Improves Complexity on a Common Problem Class

Consider objectives with initial gap at most Δ, smoothness constant at most L, and unbiased oracles satisfying BV and centered δ-similarity. Unbiasedness gives

$$\begin{aligned}\mathbb{E}[\|h(x,\xi) - h(y,\xi)\|^2] &= \|\nabla f(x) - \nabla f(y)\|^2 + \mathbb{E}[\|r(x,\xi) - r(y,\xi)\|^2] \\ &\le (L^2 + \delta^2)\|x - y\|^2.\end{aligned} \tag{E.4}$$

The mean-squared-smoothness guarantees of SPIDER and PAGE therefore apply [5, 13]: they give $O(\varepsilon^{-3})$ complexity, compared with the $O(\varepsilon^{-4})$ bounds in (E.1). Here the accuracy target is $\mathbb{E}[\|\nabla f(\widehat{x})\|] \le \varepsilon$; our squared-norm target implies it by Jensen's inequality. Both use shared-sample queries, precisely the access needed for TSGD differences.

The following specialization of Fradin et al. [6] shows that the exponent three is optimal on this same class, not merely on a larger mean-squared-smooth class. It applies to zero-respecting algorithms, which introduce only coordinates revealed by past oracle responses when started at the origin, rather than to all randomized algorithms.

Corollary E.1 (Specialization of Fradin et al., Theorem 3.2)**.** *Fix $\Delta, L, \sigma, \delta > 0$. For all sufficiently small $\varepsilon > 0$, any zero-respecting algorithm that guarantees $\mathbb{E}[\|\nabla f(\widehat{x})\|] \le \varepsilon$ uniformly over the class satisfying Assumptions A1–A4 and $f(x_0) - f_{\inf} \le \Delta$ requires*

$$\Omega\left(\frac{\Delta\delta\sigma}{\varepsilon^3}\right) \tag{E.5}$$

stochastic-gradient evaluations in the worst case. The oracle permits any fixed finite number of simultaneous query points sharing the same sample, with independent samples across rounds. The same lower bound holds for the stronger target $\mathbb{E}[\|\nabla f(\widehat{x})\|^2] \le \varepsilon^2$.

Proof. Set $p = q = 2$ in Theorem 3.2 of Fradin et al. [6]. Their bounded central moment and (q, δ)-similarity assumptions then become BV and centered δ-similarity, respectively. With $L_1 = L$ and $\sigma_1 = \sigma$, their lower bound is

$$\Omega\left(\min\left\{\frac{L\Delta}{\varepsilon^2} + \frac{L\Delta\sigma^2}{\varepsilon^4}, \quad \frac{\sigma^2 + (L+\delta)\Delta}{\varepsilon^2} + \frac{\delta\Delta\sigma}{\varepsilon^3}\right\}\right).$$

The theorem applies for $0 < \varepsilon \le c_1\sqrt{L\Delta}$ with a universal $c_1 > 0$. For fixed positive parameters, the first branch grows as $L\Delta\sigma^2/\varepsilon^4$, whereas the second is asymptotic to $\delta\Delta\sigma/\varepsilon^3$. Their ratio tends to infinity, so the minimum is the second branch for sufficiently small ε, giving the claimed bound. The lower bound counts shared-sample query rounds, each requiring at least one gradient evaluation, so it also bounds the evaluation count. Finally, $\mathbb{E}[\|\nabla f(\widehat{x})\|] \le \sqrt{\mathbb{E}[\|\nabla f(\widehat{x})\|^2]}$ extends the bound to the squared-norm criterion. □

Together with the SPIDER/PAGE upper bounds, Corollary E.1 establishes optimality of the ε^{-3} exponent among zero-respecting algorithms on this common class as $\varepsilon \to 0$, with the positive bounds $\Delta, L, \sigma, \delta$ held fixed. This does not establish an $\Omega(\varepsilon^{-4})$ lower bound for pure TSGD; its $O(\varepsilon^{-4})$ guarantee need not be tight, and exact transport can already yield faster convergence.

E.4 Error Recursions for Refreshing and Mixing

Table 3 gives the estimator identities. Here we record how these operations affect tracking error. Write $u_{k+1} := H_{k+1} - \nabla f(x_{k+1})$, $q_k := \mathbb{E}[\|d_{k+1}\|^2]$, and $v_{k+1} := \mathbb{E}[\|u_{k+1}\|^2]$. Both u_{k+1} and d_{k+1} are conditionally unbiased given $\mathcal{F}_k$.

Restarts and probabilistic refreshing. Within a transport phase starting at s, the martingale identity gives

$$E_k = E_s + \sum_{t=s}^{k-1} q_t. \tag{E.6}$$

An exact refresh resets E_s to zero; an independent batch of size B gives $E_s \le \sigma^2/B$ under BV. This limits the accumulation window without undoing the effect of past errors on the iterates. For PAGE, an independent Bernoulli decision selects a fresh estimate with probability p and transport otherwise [13]. Conditioning on that decision yields

$$E_{k+1} = (1-p)(E_k + q_k) + p\, v_{k+1}. \tag{E.7}$$

The branches are mutually exclusive, so their noises have no cross term. MARINA uses the same principle with exact refreshes and unbiased compressed gradient differences [8]. Unbiased compression ensures the conditional mean, but a similarity bound also requires control of compression variance and local gradients.

Mixing fresh and transported estimates. For MVR/STORM with deterministic fresh weight α, subtracting the true gradient from the update in Table 3 gives

$$e_{k+1} = (1-\alpha)e_k + (1-\alpha)d_{k+1} + \alpha u_{k+1}. \tag{E.8}$$

Conditional unbiasedness removes cross terms with e_k, hence

$$\begin{aligned} E_{k+1} = (1-\alpha)^2(E_k + q_k) + \alpha^2 v_{k+1} \\ + 2\alpha(1-\alpha)\mathbb{E}[\langle d_{k+1}, u_{k+1}\rangle]. \end{aligned}$$

Unlike probabilistic refreshing, mixing uses both noises together. With shared sampling their covariance cannot generally be discarded. For history-dependent weights, this identity holds conditionally; the weights remain inside expectations when averaging over the history.

Scaled transport within momentum. The unclipped MARS estimator in Table 3 interpolates between MSGD at $\gamma = 0$ and MVR at $\gamma = 1$. For $0 \le \gamma \le 1$,

$$g_{k+1}^{\mathrm{MARS}} = (1-\gamma)g_{k+1}^{\mathrm{MSGD}} + \gamma g_{k+1}^{\mathrm{MVR}},$$

where the two candidates use the same incoming estimate, sample, and fresh weight β [1]. Pure transport is the $(\beta, \gamma) = (0, 1)$ limit of this expanded identity, not a substitution into MARS's original rescaled formula, which is undefined at $\beta = 0$. A scaled difference with $\gamma \neq 1$ is generally not an unbiased estimate of the gradient change. Cached samples, clipping, and preconditioning also fall outside the pure-transport assumptions. VENUS changes the coupling of fresh and transported samples, and therefore their covariance, while preserving shared sampling within each difference [1].